

Principles of Quantum Mechanics

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1 Introduction

Recall from IB Quantum Mechanics we have the following:

- (i) Wave-particle duality. Waves behaving like particles like light quanta and vice-versa (particles behaving like waves).
- (ii) We have a wavefunction $\psi(x)$ for a particle which we interpret as a probability density $|\psi(x)|^2$ (i.e. probability is intrinsic to the theory).
- (iii) Observables become operators on wavefunctions:

- $\hat{x} \rightarrow x$;
- $\hat{p} \rightarrow -i\hbar \frac{\partial}{\partial x}$

and the lack of commutation since $[\hat{x}, \hat{p}] = i\hbar$ leads to the uncertainty relation.

- (iv) Schrödinger's equation with Hamiltonian H specifies dynamics and eigenvalues of H are the allowed energy levels of the system.
- (v) All of this explains the hydrogen atom structure and the resulting line spectra.

The aims of this course are the following:

- (i) Reformulate quantum mechanics in a more powerful, flexible and useful form: the Dirac formulation, which goes beyond wavefunctions. The previous description appears as a particular realisation of our generalisation of quantum states and also leads to alternatives such as

$$\tilde{\psi}(p) \quad \text{with } \hat{x} \rightarrow i\hbar \frac{\partial}{\partial p} \quad \text{and } \hat{p} \rightarrow p.$$

- (ii) This more general approach allows much simpler analysis of known problems (for example the Simple Harmonic Oscillator) but also lots of new things such as *spin* of particles and symmetries (translations, rotations) and the related conservation laws.
- (iii) We will also look at identical particles (bosons and fermions).
- (iv) We will develop a framework for quantising more general systems such as EM fields.
- (v) We will also setup perturbation theory which is a practical tool for computation but we will not heavily consider the detailed applications.

2 Dirac Formalism

2.1 Concepts and Definitions

A quantum system is described at each instant by a *state* $|\psi\rangle$ which belongs to a complex vector space V ; any non-zero element of this space V is a possible state. Thus

$$|\psi\rangle, |\phi\rangle \in V \implies \alpha |\psi\rangle + \beta |\phi\rangle \in V \text{ for any } \alpha, \beta \in \mathbb{C}.$$

Physically this is a superposition principle leading to wave-like behaviour but these states are *not wavefunctions*. There are also *dual* or conjugate states $\langle\psi|$ which belong to the dual space $V^\dagger$. By definition states and duals can be combined or paired to give a complex number. The map

$$\langle\psi|, |\psi\rangle \rightarrow \langle\psi|\psi\rangle$$

defines a map from $V^\dagger \times V \rightarrow \mathbb{C}$ with

$$\langle \phi | \alpha_1 | \psi_1 \rangle + \alpha_2 | \psi_2 \rangle = \alpha_1 \langle \phi | \psi_1 \rangle + \alpha_2 \langle \phi | \psi_2 \rangle$$

and

$$\langle (\beta_1 \langle \phi_1 | + \beta_2 \langle \phi_2 |) | \psi \rangle = \beta_1 \langle \phi_1 | \psi \rangle + \beta_2 \langle \phi_2 | \psi \rangle.$$

The spaces V and $V^\dagger$ comes with an inner product which can be described as a one-to-one correspondence between the spaces

$$\begin{aligned} V &\rightarrow V^\dagger \\ | \psi \rangle &\rightarrow \langle \psi | = (| \psi \rangle)^\dagger. \end{aligned}$$

We also have that if we take a linear combination of states to the dual space, the coefficients have to have their complex conjugates taken to from the dual space to V or visa-versa. Hence the inner product is

$$| \phi \rangle, | \psi \rangle \rightarrow \langle \phi | \psi \rangle = (| \phi \rangle)^\dagger | \psi \rangle$$

and it is assumed to obey

$$\langle \phi | \psi \rangle = \langle \psi | \phi \rangle^*$$

since the inner product is hermitian, and is the inner product defines a positive definite norm on V .

$$|| | \psi \rangle ||^2 = \langle \psi | \psi \rangle.$$

Remark. Knowing $\langle \phi | \psi \rangle$ for all $\langle \phi |$ determines $| \psi \rangle$ and visa-versa.

We can show this by

$$\langle \phi | \psi_1 \rangle = \langle \phi | \psi_2 \rangle \implies \langle \psi | \chi \rangle = 0$$

where $| \chi \rangle = | \psi_1 \rangle - | \psi_2 \rangle$. If this holds for all $\langle \phi |$ we can set $\langle \phi | = \langle \chi |$ so by positive definiteness we have that the norm of $| \chi \rangle$ is zero, so $| \chi \rangle = 0$ too.

We also have that the physically content of any state is unchanged by

$$| \psi \rangle \rightarrow \alpha | \psi \rangle$$

for any $\alpha \in \mathbb{C}$ non-zero. So we will usually normalise to make the state have norm 1, but we can still change by a phase vector.

Remark. Relative phases can be important in expressions where we're adding a linear combination of elements of V . We also assume the space V is complete so appropriate sequences and series converge in V .

A complete inner product space is called a *Hilbert space*.

Definition. (Operator) An *operator* Q is a linear map on states

$$| \psi \rangle \rightarrow Q | \psi \rangle.$$

The same operator can be thought of as acting to the left on dual states, so

$$\langle \phi | \rightarrow \langle \phi | Q$$

where the right hand side is defined by

$$(\langle \phi | Q) | \psi \rangle = \langle \phi | (Q | \psi \rangle) = \langle \phi | Q | \psi \rangle$$